



Assessment of Nitrate Contamination Due to Groundwater Pollution in Southeastern Part of Anantapur District, Andhra Pradesh

V. SUNITHA¹, B. MURALIDHARA REDDY¹ and M. RAMAKRISHNA REDDY²

¹Department of Geology & Geoinformatics, ²Department of Earth Sciences

Yogi Vemana University, Kadapa – 516 003

Email: vangala_sunitha@yahoo.com, bandi.murali@yahoo.co.in, reddy.mrk@rediffmail.com

Abstract: Groundwater is the major drinking water source in India, deterioration in its quality is going unchecked. Nitrate leaching is a common phenomenon. Contamination of drinking water by nitrate leads to a number of health problems to human beings and animals. In an agro-economy based rural areas, the nitrate contamination is rampant and much attention has not been drawn towards this anthropogenic pollution. In the present study area groundwater samples have been collected during pre and post monsoon seasons and analysed for the major ions such as Ca^{+2} , Mg^{+2} , Na^+ , K^+ , CO_3^{2-} , HCO_3^- , Cl^- , SO_4^{2-} , F^- and NO_3^- . The study revealed that 88 percentage of the samples were found to be unsuitable for drinking purposes in the pre monsoon season and 86 percentage in the post-monsoon season due to excess nitrate ($>45 \text{ mg/l}$) content in groundwater. Intense agricultural practices, improper sewage and organic waste disposal methods have been found to contribute to nitrate contamination in the groundwater.

Keywords: Southeastern part of Anantapur, groundwater, nitrate, fertilizers

Introduction:

Nitrogen, an element considered to be the most abundant in the atmosphere, comprising nearly 80% (Berner and Berner, 1987) can be found in many forms, the major ions being N_2 , N_2O , NO , NO_2 , NH_3 (Gaillard, 1995). Nitrate is part of the nitrogen cycle in nature and represents the most oxidized chemical form of nitrogen found in the natural systems. All living systems need nitrogen for their existence since it is used to build many essential components such as proteins, DNA, RNA, Vitamins and as well as hormones and enzymes. Nitrogen though very essential for the very existence of life, is also one of the most widespread pollutants of groundwater in many parts of the world and in several instances this is due to the intensification of agriculture (Goldberg V. M., 1989).

The groundwater pollution due to nitrates is increasing in India. The water quality assessment studies carried out in 17 Indian states by NEERI showed that out of 4,696 water samples, 1,290 samples (27%) have nitrate exceeding the drinking water standard (Bulusu and Pande, 1990). The nitrate concentration in groundwaters has shown rising trends in many countries in the last 30 years (Guarda et al., 2004). Application of nitrogen based fertilizers such as NPK (Nitrogen, Phosphorous, Potassium), urea together with organic manure like cow dung, decomposed vegetative waste, in more than required quantities, could lead to the leaching and percolation of nitrate in to sub-surface water bodies.

Improper disposal of the human and animal waste, unlined drainage and sewerage lines may also add to the nitrate contamination of groundwater (Jack and Sharma, 1983). Agriculture and livestock production accounts for nearly 80% of all the nitrogen added to the environment. Nearly 40% of the fertilizers in India are used for paddy cultivation and about 50% of applied nitrogen fertilizers are leached out and escape into the environment (Punjab Environment Information, 2005).

Nitrates as a Health Hazard:

In India, high concentration of groundwater nitrate (more than 45 ppm) have been found in many districts of Andhra Pradesh, Bihar, Delhi, Haryana, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Tamilnadu, Rajasthan, West Bengal and Uttar Pradesh (Patil et al., 2009). The highest groundwater nitrate concentration of 3080 ppm was found in Bikaner in Rajasthan (C.G.W.B). The study carried out by (A.G.S Reddy et al., 2007) in Anantapur district revealed that 65% of samples have shown nitrate levels above the desirable limit (45 ppm). Sources of nitrate in groundwater can be considered in four categories: (1) Natural sources (2) waste materials (3) irrigated agriculture and (4) row crop agriculture. The major sources include intensive animal operations with nitrate from over application of animal wastes, and irrigated and row crop agriculture, with nitrate from fertilizer-induced mineralization of soil organic nitrogen and from over application. Septic tank system and other

sources, such as landfills can be concern in localized areas (Canter, 1996). High concentration of nitrate in the drinking water causes many serious health problems especially in children and young livestock. Due to its hazardous nature, the permissible limit of nitrate for drinking water has been reduced from 100 mg/l to 45 mg/l by the Bureau of Indian standards in recent years. Although methemoglobinemia (blue-baby syndrome) is the most immediate life threatening effect of the nitrate exposure, there are a number of equally serious long terms, chronic impacts. In numerous studies, exposure of high levels of nitrate in the drinking water has been linked to a variety of effects ranging from enlargement of the thyroid to 15 types of cancer and two kinds of birth defects and even hypertension (Forman et al., 1985). Research shows a definite relationship between increasing rates of stomach cancer with increasing nitrate intake (Payne, 1993). Through the effect of continuous consumption of high nitrate in water may not physically visible as in the case of fluoride, the local people have complained of recurring indigestion and gastrointestinal problems in the study area.

Study Area:

The study area is located in southeastern part of Anantapur district, Andhra Pradesh and lies between longitudes $77^{\circ} 30' - 78^{\circ} 15'$ East and latitudes $14^{\circ} 0' - 14^{\circ} 30'$ North and falls in the Survey of India toposheet Nos. 57 F/14, F/15, J/3, J/4 as shown in Fig.1. The study area is mainly covered by peninsular gneisses of Archaean age consisting of pink granites, schists, composite gneisses of Dharwar age, intruded by a few pegmatite dykes and numerous dolerite dykes and the possible diamondiferous volcanic pipes. The geological map of the study area is shown in the Fig.2. Geomorphic features observed from the study area are denudational hills, dissected pedimonts, pediplain, flood plain. The climate is semiarid with mean monthly temperatures of 15°C in January and 39°C in May. The normal rainfall of the district is 553 mm of which it secures the least rainfall as compared to Rayalaseema and other parts of Andhra Pradesh.

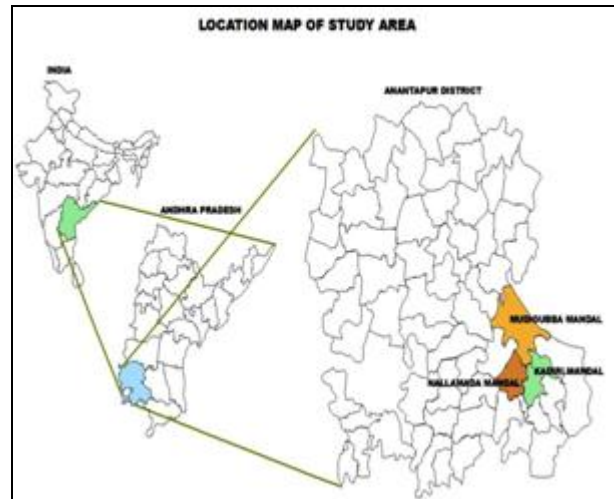


Figure 1: Location Map of the Study Area

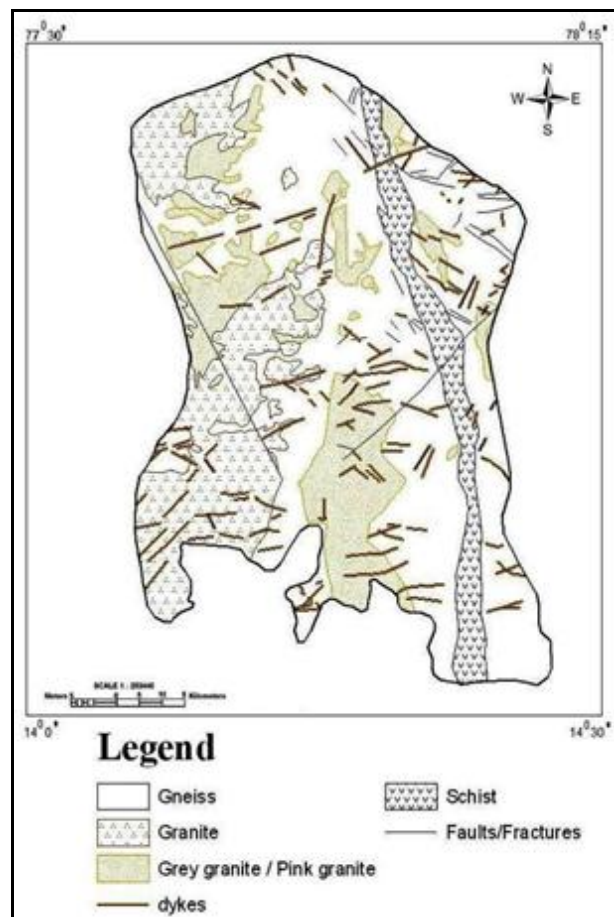


Figure 2: Geological Map of the Study Area

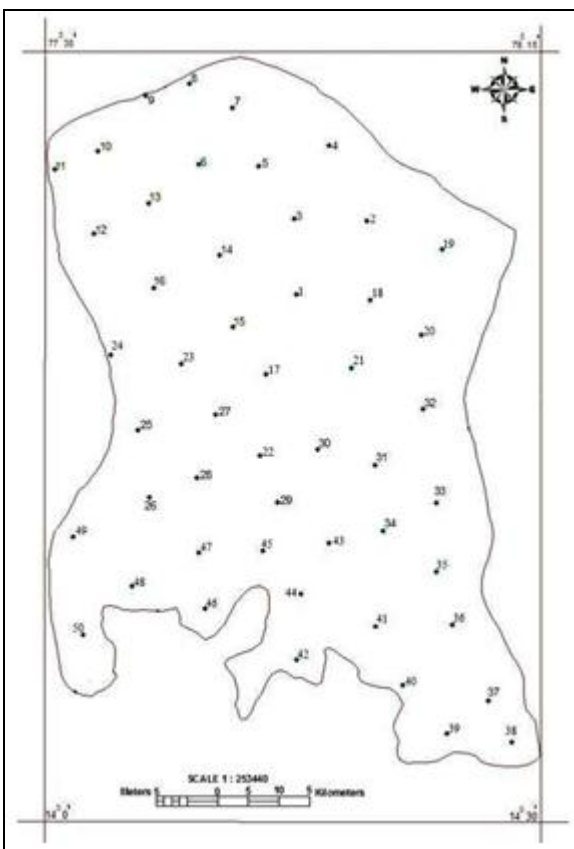


Figure 3: Sample Location Map of the Study Area

Table 1: Statistical Parameters of the Different Chemical Constituents of Groundwater in Study Area Pre-Monsoon Season and Post-Monsoon (mg/l)

Pre-monsoon					Post-Monsoon				
Parameters	Minimum	Maximum	Mean	Standard Deviation	Parameters	Minimum	Maximum	Mean	Standard Deviation
pH	7.0	8.6	8	0.40	pH	7.0	8.6	8	0.40
Ec	310	2580	732.6	164.3	Ec	280	2500	659.6	394.18
Ca ²⁺	70	360	76.42	615.2	Ca ²⁺	50	300	145.19	62.45
Mg ⁺²	30	180	67.82	32.49	Mg ⁺²	28	280	74.00	45.69
Na ⁺	280	850	192.16	168.24	Na ⁺	260	840	578.46	199.18
K ⁺	30	180	211.98	32.36	K ⁺	28	150	59.92	28.49
SO ₄ ⁻²	80	310	186.68	60.80	SO ₄ ⁻²	73	300	178.01	67.04
Cl ⁻	72	480	1629.2	102.86	Cl ⁻	70	460	187.84	100.27
HCO ₃	75	320	676.44	59.11	HCO ₃	75	300	168.92	65.24
TDS	650	3280	3.46	793.01	TDS	620	3200	1431.7	757.90
TH	246	1474	65.94	256.85	TH	456	1898	716.13	323.35
F	1.1	7.1		1.21	F	1.1	7.1	3.46	1.21
NO ₃	20	110		21.02	NO ₃	20	100	61.67	22.96

The groundwater nitrate concentration was found in the range of 20 to 110 mg/l and 20 to 100 mg/l for the month of April and October 2010 respectively. Maximum groundwater nitrate concentration in the study area was found to be (110 mg/l) Malakavemula Cheruvu in pre-monsoon and minimum groundwater nitrate concentration Of (20mg/l) was found in Nagurivandlapalli. In the command areas intense paddy cultivation was noticed where more than the required

Materials and Methods:

Groundwater samples were collected from bore wells of the rural water supply representing different hydrogeological environments (Fig. 3) water samples were collected following standard (APHA) methods. The temperature pH and EC were measured at the time of sampling. Samples were subsequently analysed as per the APHA (1992) procedures. Fluoride and nitrate were analysed by Ion-selective electrode (Model: pH/ISE), the results of which were tested for accuracy with the charge balance error method (Domenico and Schwartz, 1990).

Results and Discussion:

The groundwater of the study area is slightly alkaline as its pH varies from 7 to 8.6 mg/l in the pre monsoon season. The EC range from 310 to 2580 μ S/cm with a mean of 732.6. The average total hardness (as CaCO₃) is 676.4 mg/l with the minimum being 246 and maximum 1474 mg/l. The Na⁺, K⁺ and Cl⁻ are found to be high in the pre-monsoon samples. EC, TH, TDS, Mg⁺² and F⁻ concentrations are found to be more than the permissible limits of Bureau of Indian standards (BIS 2003).The statistical analysis of various chemical parameters is shown in table 1.

quantity of water, nitrogen based fertilizers, manure and pesticides are used extensively which all contribute to the non point source contamination of nitrates in ground water. Another favorable factor for high nitrate accumulation in command areas is that they are underlain by granites which facilitate more leaching of nitrate to ground water.

Table 2: Nitrate Concentration of the Study Area in Pre-Monsoon and Post- Monsoon

Percentage of Samples with NO ₃ (mg/l)			
	< 45 mg/l	45-100 mg/l	>100 mg/l
Pre-monsoon	12%	86%	2%
Post-monsoon	14%	86%	Nil

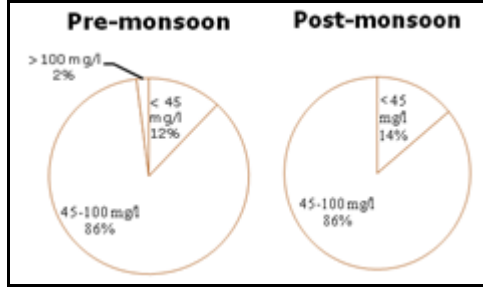


Figure 4: Pie Diagram Represents Nitrate Concentrations

The content of many chemical constituents are low (Table 4) indicating a decreased mineralization in the post-monsoon season due to dilution by addition of rain or irrigation water to the aquifer. High concentration of nitrate is more prevalent in the granitic terrain,

command areas and in discharge zones in the pre monsoon season, only 12% of the samples have nitrate less than 45 mg/l, 86% of the samples between 45-100 mg/l and 2% of the samples have nitrate content of more than 100 mg/l (Table 2) and pie diagrams represents Nitrate concentrations (Fig. 4).

Table 3 gives details about the crops under cultivation, average and fertiliser dosage to the crop in southeastern part of Anantapur District, A.P. Maximum amount of farm yard manure and chemical fertilisers is used for Paddy when compared to other crops. The quantity of fertiliser used for Paddy is more than 10 times that used for other crops. This fact substantiates the conclusion that Paddy cultivation is responsible for the high nitrate content of surrounding ground water sources. The nitrogen based fertilizers or manure used on a sandy soil are more vulnerable to leaching into groundwater than the nitrogen fertilizers used on a clay soil; as water moves rapidly through sandy or coarse textured soils (Bhumbla, 2006; Voudouris et al., 2004). Highly weathered mantle and good network of fractures in granites down to 30-40 m might have facilitated downward percolation of nitrate leading to its enhancement in shallow and moderately deep aquifers.

Table 3: Agricultural Practices in Study Area

S. No	Crops under Cultivation	Area in Hectares	Fertiliser dosage recommended Kg/Hectare			Farm you manure
			N	P	K	
1	Paddy irrigated	3988	100-120	75	30	500
2	Jowar	1407	80-100	40	40	500
3	Bajra	577	100	50	30	-
4	Red gram	191	20	40	-	-
5	Castor	448	40	20-30	20	-
6	Ragi	381	40	40	40	-
7	Vegetables & others	600	25	50	-	-

Ground waters at Malakavemula cheruvu, Uppalapadu, Vaddevaripalli, Nakkalaguttapalli, Seelamvaripalli, Pathabathalapalli, have nitrate values more than 100 mg/l. minimum value of nitrate concentration in ground water is 20 mg/l at Nagurivandlapalli, Pandulakunta. In increased nitrate concentrations at low, in more number of samples is attributed to the pollution from fertilizers. However, high concentrations of nitrate observed at few places are attributed to other sources rather than fertilizer. A thorough observation at the wells, where high concentrations of nitrate are observed, indicated that there is an association between high nitrate concentrations and animal wastes at cattlesheds which are located near the well sites. The animal wastes at cattlesheds acted as point sources. High concentrations of nitrate are also observed in the groundwater nearer to

the markets where animal trading was talking place once in a week.

As depicted in table 4 the chemical composition of groundwater in the study area, it has been observed that the groundwater of shallow aquifer has a slightly lower groundwater nitrate concentration than the deeper aquifer. With respect to the seasonal variation of groundwater nitrate concentration in the study area it was observed that the ground water nitrate concentration in pre-monsoon season is higher than the post-monsoon. Evapotranspiration leading to accumulation soils in ground water and the non availability of recharge water are some of the reasons for higher nitrate content in groundwater during the pre monsoon. The details of the nitrate sampling sites are presented in Table 4.

Table 4: Sampling Locations of the Study Area in Pre-Monsoon Season and Post-Monsoon Season (mg/l)

S. No.	Names of the Villages	Nitrate in mg/l	Nitrate in mg/l	Depth of Wells
		Pre-monsoon	Post-monsoon	
1	Ralla Anantapuram	58	54	50
2	krishnapuram	55	50	200
3	Nallacherlopalli	60	58	140
4	Yerravankapalli	68	65	150
5	Gunjepalli	90	85	200
6	Jonnalakothapalli	92	92	80
7	Muktapuram	70	70	50
8	Krishnapuram	30	28	60
9	Bapanakunta	62	60	150
10	Reddipalli	45	40	20
11	Vankarakunta	48	45	90
12	Nallamada	56	50	80
13	Pullagurapalli	70	68	85
14	Toletivaripalli	72	70	250
15	Donnekota	56	54	60
16	Sankepalli	48	40	70
17	Ramaswamy Tanda	98	95	10
18	Sanevaripalli	78	70	20
19	Rachavaripalli Tanda	65	65	50
20	Malakavemula	60	60	220
21	Nagurivandla Palli	20	78	60
22	Alampur	70	70	300
23	Vaddevari Palli	100	95	150
24	K.Kuntlapalli	46	45	70
25	Kondagattu Palli	56	50	140
26	Chenchugaripalli Tanda	90	85	100
27	Seelamvari Palli	97	90	210
28	Ramireddi Palli	88	85	120
29	Payagattu Palli	70	70	30
30	Yerikireddi Palli	66	65	300
31	Pandulakunta	20	20	220
32	Kothannagari Palli	68	65	250
33	Yerravanka Palli	49	45	350
34	Podaralla Palli	56	50	310
35	Timmanayana Palem	70	68	280
36	Mallepalli	77	70	50
37	Tappetavari Palli	56	50	50
38	Chiragarapalli	72	70	90
39	Guttakindapalli Tanda	54	52	120
40	Gollapalli	80	75	140
41	Eddulavaripalli Tanda	80	75	200
42	Choutakuntapalli	63	60	250
43	Kothapalli	33	30	320
44	R.Ramapuram	35	35	250
45	Chandrayani Palli	32	30	250
46	Pathabathala Palli	90	85	220
47	Uppalapadu	100	90	180
48	MalakavemulaCheruvu	110	95	250
49	Nakkalaguttapalli	90	100	220
50	Aakuthotapalli	78	90	300

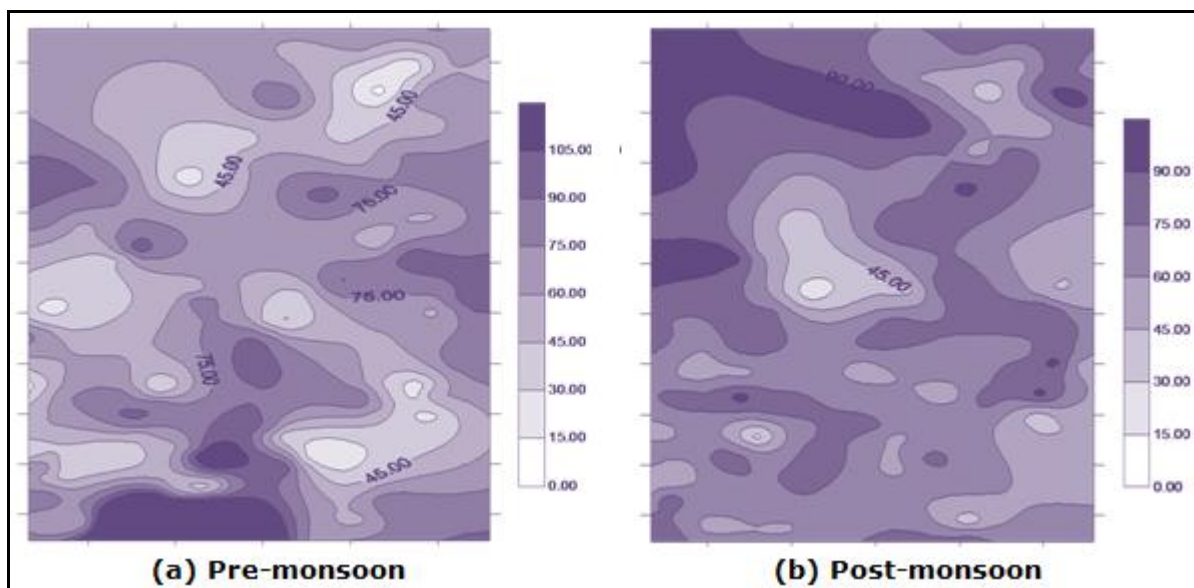


Figure 5: Nitrate Distribution Maps of the Study Area

The concentration of nitrate ranges from 20 mg/l to 110 mg/l in pre monsoon and 20 mg/l to 100 mg/l in post monsoon; the high concentration of NO_3^- is attributed to increased usage of fertilizers, decaying organic matter; the large variation is attributed to contaminant by sewage wastes. Distribution maps of nitrate during pre and post monsoon seasons is shown in Fig. 5.

Conclusion:

The study revealed that 88 percentage of the samples were found to be unsuitable for drinking purposes in the pre monsoon season and 86 percentage in the post-monsoon due to excess nitrate (>45 mg/l) content in groundwater. Fertilizers and organic manure are the major sources; animal waste dumps and domestic sewerage are minor source of nitrogen contamination to groundwater in the study area. The most effective way of reducing the nitrogen content of groundwater in the areas where agriculture is the main occupation is by reducing the application of fertilizers in consultation with agriculture scientists and by changing the cropping pattern by going in for irrigated dry crops which consume less water and fertilizers and frequently modifying the cropping sequences which offer possible ways to scavenge the nitrogen.

Acknowledgements:

The work was funded by the Department of Science and Technology (DST), New Delhi under the Fast track Young Scientist scheme (Project no: SR/FTP/ES-50/2007). The financial support from DST for carrying out this study is gratefully acknowledged. The useful comments of anonymous reviewers are appreciated.

References:

- [1] A.G.S.Reddy, K.Niranjan Kumar, D.Subba Rao and S.Sambashiva Rao (2007). Assessment of nitrate contamination due to groundwater pollution in north eastern part of Anantapur District, A.P. India. Journal of Environ. Monit. Assess, DOI 10.1007/s10661-008-0176-y.
- [2] APHA (1992). Standard methods for the examination of water and wastewater, American Public Health Association, Washington DC.
- [3] Berner E, and Berner R., (1987). The global water cycle (pp.102-119). New Jersey: Prentice Hall.
- [4] Bhumbala D.K. (2006) Agriculture practices and nitrate pollution of water cited in website of West Virginia University Extension Service-soil Management.
- [5] BIS (2003). Indian standard specifications for drinking water (p.10). New Delhi, India: IS: 10500, Bureau of Indian Standards.
- [6] Bulusu K.R., Pande S.P., (1990). Nitrates a serious threat to groundwater pollution. Bhu-Jal News 5:39-43
- [7] Canter L.W., (1996). Nitrates in groundwater. CRC Press. ISBN 0873715691.
- [8] CGWB (2003). Results of ground water exploration in Anantapur District, AP Unpublished Reports of Central Groundwater board.
- [9] Deepika G. Patil., A.G. Takarkhede and R.K. Kamble (2009). Groundwater Nitrate in Handrapur city in Maharashtra, Indian Jour of Environ Protec., 29(8):734-738 (2009).
- [10] Domencio P.A. and Schwartz F.W., (1990). Physical and chemical Hydrogeology, New York: Wiley.

- [11] Even Paula F., and Iqbal Mohammad Z., (2005). Nitrate pollution ground water in Gilbertville, Iowa, North central section-39th annual meeting paper No 42-2 Geological society of America Abstracts with programe vol.37 No. 5, p 93.
- [12] Forman D., Al-Dabbagh S., & Doll, R., (1985). Nitrates, nitrites and gastric cancer in Great Britain. *Nature*, 313, 620-625.
- [13] Gaillard, J.F., (1995). Lecture on Nitrogen cycle as referred in Nitrogen pollution of groundwater- prepared by Lee Haller et al.
- [14] Goldberg, V.M., (1989). Ground water Pollution by nitrates from livestock wastes. *Environ Health Perspect*, 83, 25-29.
- [15] Guarda G., Padovan S. and Delogu G., *Eur. J. Agron.* (2004), 21, 181-192.
- [16] Jack G., and Sharma V.P. (1983). Nitrogen circulation and nitrate in ground water in an agriculture catchment in Southern India. *Environmental Geochemistry*, 5(2), 61-64.
- [17] Payne M.R. (1993). Farm waste and nitrate pollution. In J.G. Jones (Ed.), *Agriculture and the Environment* (pp. 63-73). New York: Ellis Horwood Limited.
- [18] Spruill T.B., Eimers J.L., and Morey A.E. (1996). Nitrate-Nitrogen concentration in shallow ground water of the coastal plain of the Albemarle-Pamlico drainage study unit, North Carolina and Virginia. Fact sheet 241-96, USGS.
- [19] Voudouris K., Panagopoulos A., and Koumantakis I.(2004). Nitrate Pollution in the coastal aquifer system of the Korinthos Prefecture (Greece). *Global NEST: The International Journal*, 6(1), 31-39